

Chemical Kinetics

Types of Reaction

- Very fast Reaction $\rightarrow$ complete within fraction of seconds.
eg:- ionic rxn, Acid-base rxn
- Very Slow Reaction $\rightarrow$ Requires years of completion.
eg:- Rusting of Iron (corrosion)
- Moderate Rxn $\rightarrow$ Requires finite time for completion. eg $\rightarrow$ Molecular rxn.

Rate of Reaction

$$\text{Rate} = \frac{\Delta C}{\Delta t} \rightarrow \begin{array}{l} \text{Change in conc. of Reactant} \\ \text{or product} \\ \text{change in time.} \end{array}$$

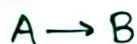
for a rxn, $aA + bB \rightarrow xX + yY$

$$\text{Rate} = -\frac{1}{a} \frac{d[A]}{dt} = -\frac{1}{b} \frac{d[B]}{dt} = \frac{1}{x} \frac{d[X]}{dt} = \frac{1}{y} \frac{d[Y]}{dt}$$

Unit of Rxn:-

- for liquid rxn $\rightarrow \text{mol L}^{-1} \text{s}^{-1}$
- for gaseous rxn $\rightarrow \text{atm s}^{-1}$

Average rate of Reaction



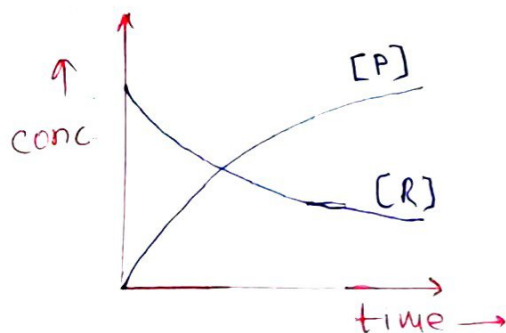
$$\text{Rate} = \frac{\text{higher value} - \text{lower value}}{\Delta t}$$

$$\text{Rate} = -\frac{\Delta[R]}{\Delta t} = -\frac{[R_2] - [R_1]}{t_2 - t_1}$$

Instantaneous rate of Rxn



$$\text{Inst. Rate} = -\frac{d[A]}{dt} = +\frac{d[B]}{dt}$$



$\Rightarrow$ Note:-

- Always take mole ratio not mass ratio for calculation of rate.
- & + shows decrease & increase concⁿ time respectively
- overall rate is +ve.
- Rate of rxn $\propto$ stoichiometric coefficient.

factors affecting Rate of rxn

- $\rightarrow$ Concentration of Reactants
- $\rightarrow$ Pressure
- $\rightarrow$ Temperature
- $\rightarrow$ catalyst

Effect on rxn rate increases with increase in factors.

Rate Law:-



$$\text{Rate} \propto [A][B]$$

$$\text{Rate} = k[A]^a[B]^b \text{ (law of mass action)}$$

$$\text{Rate} = k[A]^x[B]^y \text{ (Rate law expression)}$$

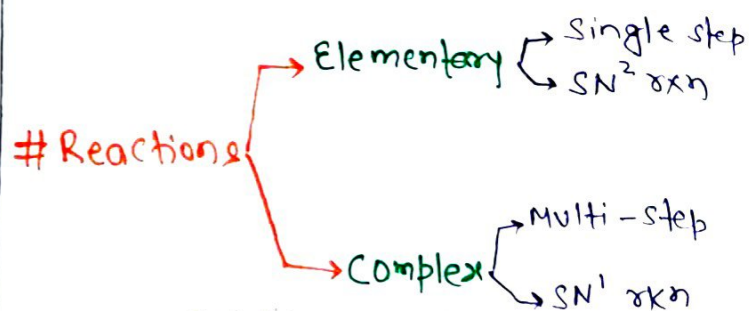
• Rate constant [K] :-

- $\rightarrow$ Larger the value of K, faster is the reaction.
- $\rightarrow$ The value of 'K' only depends on Temperature.

$$\text{Unit of 'K'} = (\text{mol})^{1-n} \text{L}^{n-1} \text{s}^{-1}$$

$n = \text{order of rxn}$

Reactions



Molecularity

- (i) It is a theoretical concept.
- (ii) It is the no. of atoms, ions or molecules that must collide simultaneously with one another to result into a chemical rxn.
- (iii) Molecularity is equal to its order.
- (iv) It is applicable only for elementary reaction.
- (v) It can be integer (1, 2, 3) but not be zero, -ve or fraction.
- (vi) Molecularity $\rightarrow$ 1, 2, 3 only.

Order of Reaction



$$\text{Rate} = k[A]^x[B]^y$$

$$\text{Molecularity} = a + b$$

$$\text{order} = x + y$$

- (i) Experimental quantity (except elementary rxn)
- (ii) Can be 0, 1, 2, 3 or (-ve) or fraction.
- (iii) Applicable for both elementary & complex reaction.
- (iv) For elementary rxn;
molecularity = order of rxn.
- (v) For complex rxn, order is given by slowest step.

order of rate constant (k)

Rxn order unit of rate constant

zero $\rightarrow$ $\text{mol L}^{-1} \text{s}^{-1}$

first $\rightarrow$ s^{-1}

second $\rightarrow$ $\text{mol}^{-1} \text{L s}^{-1}$

Calculated by: $-(\text{mol})^{1-n} \text{L}^{n-1} \text{s}^{-1}$

where, $n = \text{order of rxn}$.

Zero order Reaction

$\rightarrow$ Differential rate equation

$$\text{Rate} = -\frac{d[A]}{dt} = k[A]^0 = k$$

- Rate of zero order rxn is constant and doesn't depend on conc. of reactant.

$\rightarrow$ Integrated rate equation

$$A_t = A_0 - kt$$

$A_0 \rightarrow$ Initial conc.
 $A_t \rightarrow$ conc. at time t sec

$$k = \frac{A_0 - A_t}{t} = \frac{x}{t}$$

$k \rightarrow$ Rate constant.

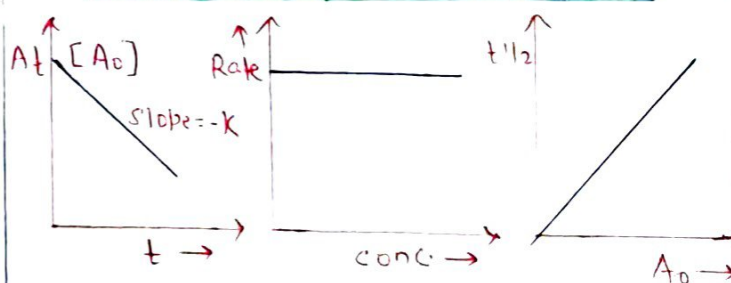
$\rightarrow$ Half life ($t_{1/2}$ or $t_{50\%}$):-

$$A_t = \frac{A_0}{2} \quad t_{1/2} = \frac{A_0}{2k}$$

$\rightarrow$ Completion time ($t_{100\%}$):-

$$A_t = 0 \quad t_{100\%} = \frac{A_0}{k} \quad [2t_{1/2} = \frac{A_0}{k}]$$

$\Rightarrow$ Graph for zero order rxn



First order Reaction

$$\text{Rate} = k[A]^1$$

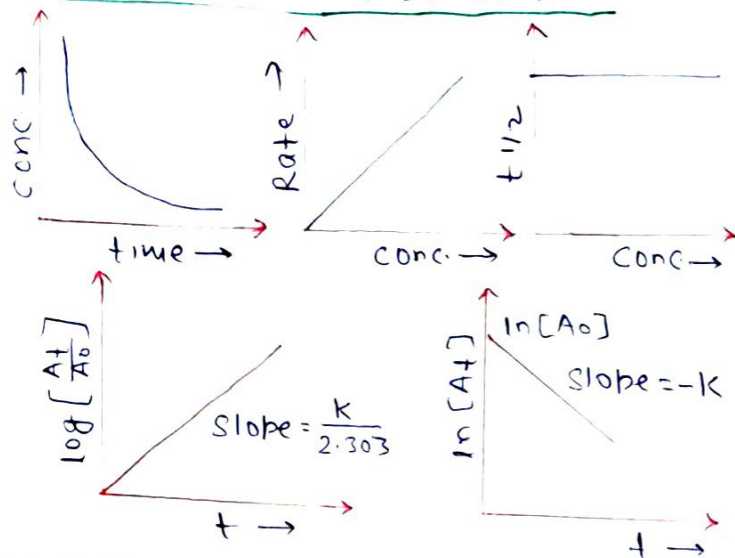
$$k = \frac{1}{t} \ln \left(\frac{A_0}{A_t} \right) = \frac{2.303}{t} \log \left(\frac{A_0}{A_t} \right)$$

$$t_{1/2} = \frac{\ln 2}{k} = \frac{2.303 \log 2}{k} = \frac{0.693}{k}$$

$\rightarrow$ [Independent of A_0]

$$\begin{aligned} t_{75\%} \text{ or } t_{3/4} &= 2t_{1/2} \\ t_{87.5\%} \text{ or } t_{7/8} &= 3t_{1/2} \\ t_{93.75\%} &= 4t_{1/2} \\ t_{99.99\%} &= 10t_{1/2} \end{aligned}$$

⇒ Graph of first order Rxn

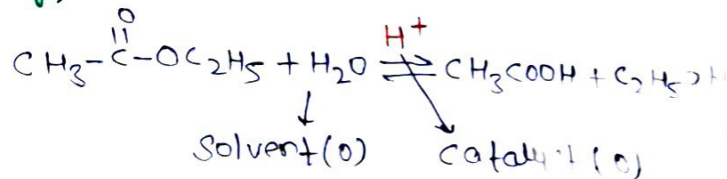


Pseudo First order Reaction

→ More than 1 reactant present but order of reactants counted w.r.t one reactant.

→ Order with respect to catalyst, solvent (H₂O) & reagent → Zero

eg:-



$$\text{order of rxn} = 1 + 0 + 0 = 1$$

eg:- Hydrolysis of ethyl acetate in acidic medium.

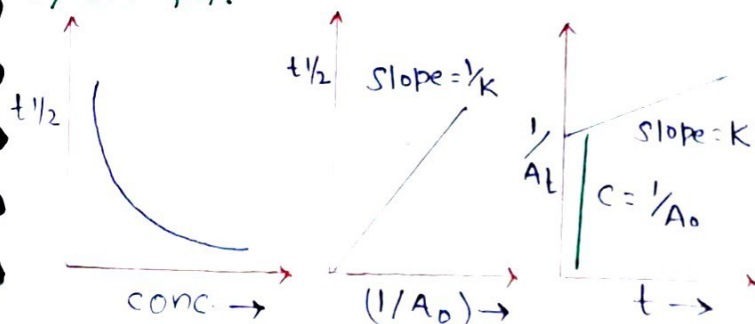
→ Inversion of cane sugar in acidic medium.

Second order Reaction

$$\text{Rate} = k[A_t]^2$$

$$k = \frac{1}{t} \left[\frac{1}{A_t} - \frac{1}{A_0} \right] \quad t_{1/2} = \frac{1}{kA_0}$$

⇒ Graph:-



nth order Reaction

$$R = k[A]^n \quad n = \text{order}$$

$$k = \frac{1}{(n-1)t} \left[\frac{1}{(A_t)^{n-1}} - \frac{1}{(A_0)^{n-1}} \right]$$

→ Valid for $n \geq 2$

$$t_{1/2} \propto [A_0]^{1-n}$$

$$t_{1/2} \propto \frac{1}{[A_0]^{n-1}} \quad \frac{(t_{1/2})_1}{(t_{1/2})_2} = \left(\frac{A_2}{A_1} \right)^{n-1}$$

Collision theory

→ Reaction occurs due to collision of molecule.

→ Rate of rxn = (orientation factor) × (energy factor)

$$\text{Rate} = PZ_{AB}e^{-E_a/RT}$$

• P is steric factor

• Z_{AB} is collision frequency of reactant A & B

Collision frequency:-

→ Total no. of collision per second per unit volume.

factors affecting rate of rxn:-

(a) Nature of Reactant & product:-

- ROR:- Gas > Liquid > Solid
- ROR:- Homogeneous > Heterogeneous
- ROR ∝ surface area
- ROR ∝ $\frac{\text{stability of product}}{\text{stability of reactant}}$

(b) Concⁿ of Reactant:-

$$R = k[\text{conc}]^n$$

where 'k' only depends on Temp.

Effect of catalyst on E_a

(a) In presence of catalyst $E_a \downarrow$ i.e.

$$\Delta E = E_a - E_p$$

(b) ΔH doesn't depend on presence of catalyst.

$$(c) \quad \frac{k_p}{k_a} = \frac{R_p}{R_a} = \frac{A_p}{A_a} e^{\Delta E / RT}$$

$k_p \rightarrow k$ in presence of catalyst
 $k_a \rightarrow k$ in absence of catalyst

(d) Temperature coefficient:-

$$\mu_T = \frac{K_{T+10}}{K_T} \approx 2 \text{ to } 3$$

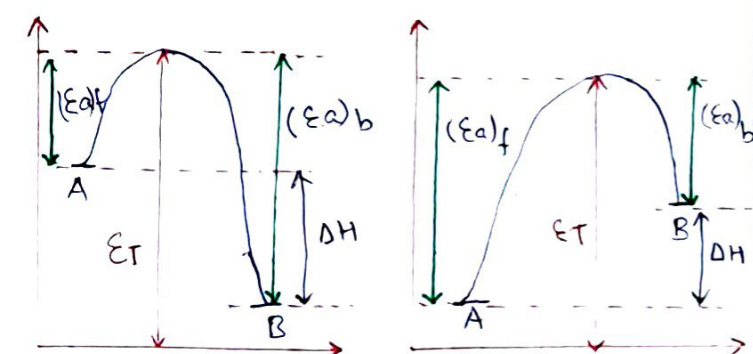
$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303R} \left(\frac{T_2 - T_1}{T_1 T_2} \right)$$

Activation Energy:-

• $E_a \rightarrow$ Energy required for reactant to be activation complex.

$\rightarrow E_a = +ve$ (always) $E_T = E_a + E_R$

$$\Delta H = (E_a)_f - (E_a)_b$$



Exothermic
(Energy released)

$$\Delta H = -ve$$

$$(E_a)_b > (E_a)_f$$

Endothermic
(Energy absorbed)

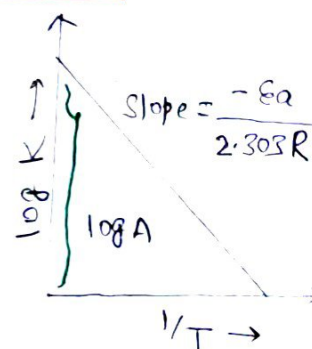
$$\Delta H = +ve$$

$$(E_a)_b < (E_a)_f$$

Arrhenius Equation

$$K = A e^{\frac{-E_a}{RT}}$$

$$\log k = \log A - \frac{E_a}{2.303RT}$$



$$T \uparrow = K \uparrow = \text{Rate} \uparrow$$

A = Arrhenius constant

E_a = Activation Energy

R = universal gas constant

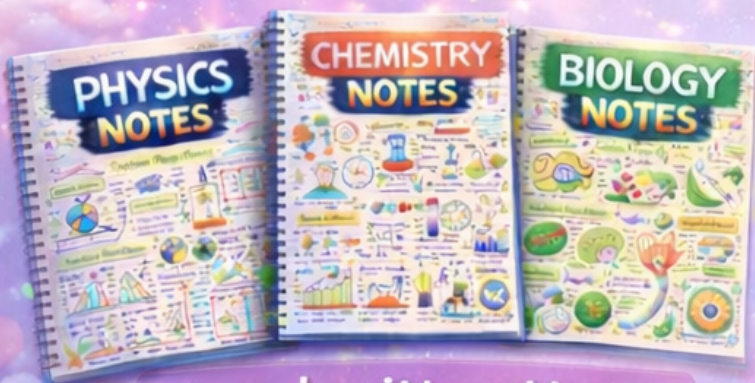
T = absolute temperature

NOTE:-

- For every 10° rise in temp., rate becomes double & hence, rate constant becomes double.
- A reaction with higher value of E_a will have smaller value of rate constant.
- Very low change in E_a occurs in very high change of Temp.

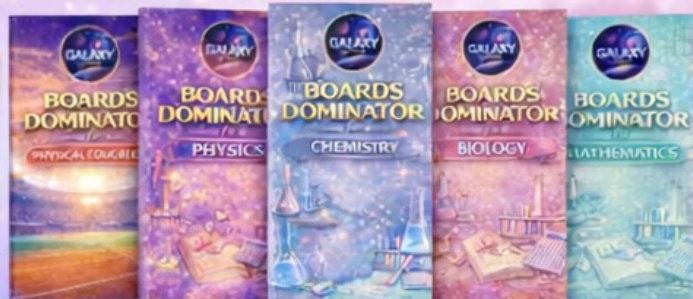
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